

CSE4203: Computer Graphics

Course Outline

Outline

- Course Objective
- Required Books
- Week-wise Plan
- Tentative Date of Quizzes
- Marks Distribution
- Some Questions

Course Objectives

- Comprehend the basics of computer graphics and the graphics pipeline
- Apply 2D and 3D linear transformation, viewing, rasterization techniques, shading models and textures.
- Analyze different algorithms used in computer graphics.

Required Books

- No specific book will be strictly followed in this course.
- Several books and online resources will be followed.
- I will share the references/book chapters/links for deeper understanding

Required Books

1. Fundamentals of Computer Graphics”, 4th Edition. Authored by: Steve Marschner Peter Shirley
2. Computer Graphics, C Version, 2nd Edition Authored by: Donald Hearn, M. Pauline Baker
3. Real-Time Rendering, Fourth Edition Authored by: Tomas Akenine-Möller, Eric Haines

Week-wise Plan

Week	Topic
1	Course Outline, Introduction to Computer Graphics and Motivation Behind the Course, Definition of Computer Graphics, Graphics Areas and Applications, Overview of Graphics API and Pipelines
2	Raster Device, Digital image representation and compression techniques, Color models and Color spaces, Raster Devices (LED and LCD)

Week-wise Plan

Week	Topic
3	Introduction to Vector Graphics, Line and curve representations, Bezier Curves Spline Curves
4	Transformation and 2D Linear Transformation, Composite Transformation, Inverse Transformations, 3D Transformation, Rotation about an arbitrary line

Week-wise Plan

Week	Topic
5	Viewing Transformation, Viewport Transformation, Orthographic Projection Transformation, and Camera Transformation, Perspective Projection
6	Rasterization, Graphics Pipeline, Bresenham's Line Drawing Algorithm, Barycentric Interpolation, Rasterizing a triangle, Antialiasing techniques, Clipping algorithms, and Hidden surface removal algorithms
7	Texture mapping, Texture Coordinate Functions, Introduction to Surface Shading, Diffuse Shading, Phong Shading, Artistic Shading

Week-wise Plan

Week	Topic
8	Introduction to Fractals and Self-Similarity, Fractal Dimension, Recursive generators, L-systems (Lindermeyer), Mandelbrot equation, Applications of fractals in computer graphics
9	Overview of Ray Tracing, Basic Ray Tracing Algorithm, Ray Sphere Intersection Ray Triangle Intersection

Week-wise Plan

Week	Topic
10	Principle of animation, Applications in Computer Graphics Keyframing, Interpolation Kinematics, Inverse Kinematics
11	Introduction to VR and AR, Devices and Sensors APIs and Hardware, Rendering Techniques Applications of VR and AR
12	Particle Motion, Ordinary Differential Equations (ODE), Particle Dynamics, Rigid-Body Dynamics Collisions

Week-wise Plan

Week	Topic
13	Introduction to machine learning and its applications in computer graphics, Deep learning-based image synthesis techniques: GANs and Variational Autoencoders
14	Traditional Object tracking Techniques, Deep Learning-Based Object Tracking, Application of Object Tracking, Review Class

Tentative Date for Quizzes

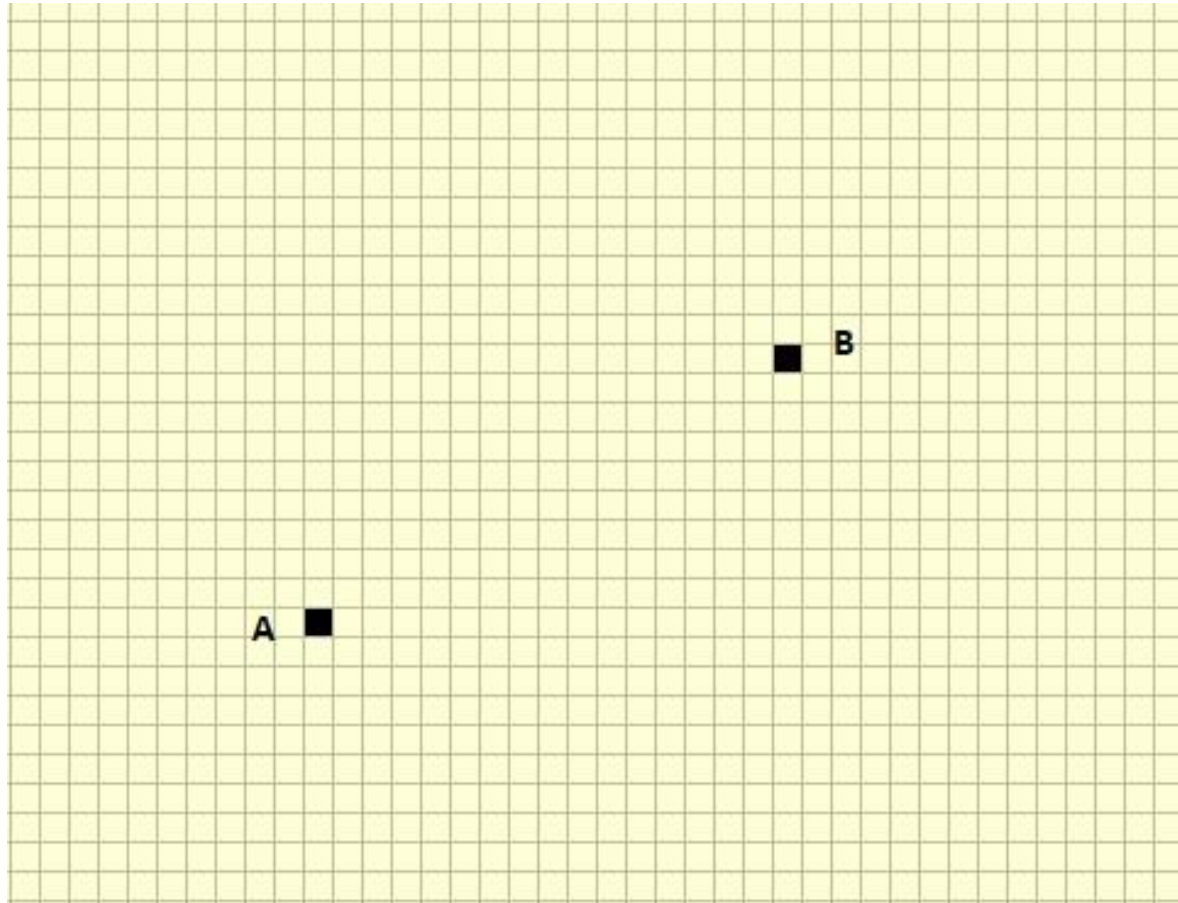
- Quiz 1 (Week - 4)
- Quiz 2 (Week - 7)
- Quiz 3 (Week - 10)
- Quiz 4 (Week - 13)

Marks Distribution

Class Performance	10%
Quizzes (Best 3 Out of 4)	20%
Final Examination	70%

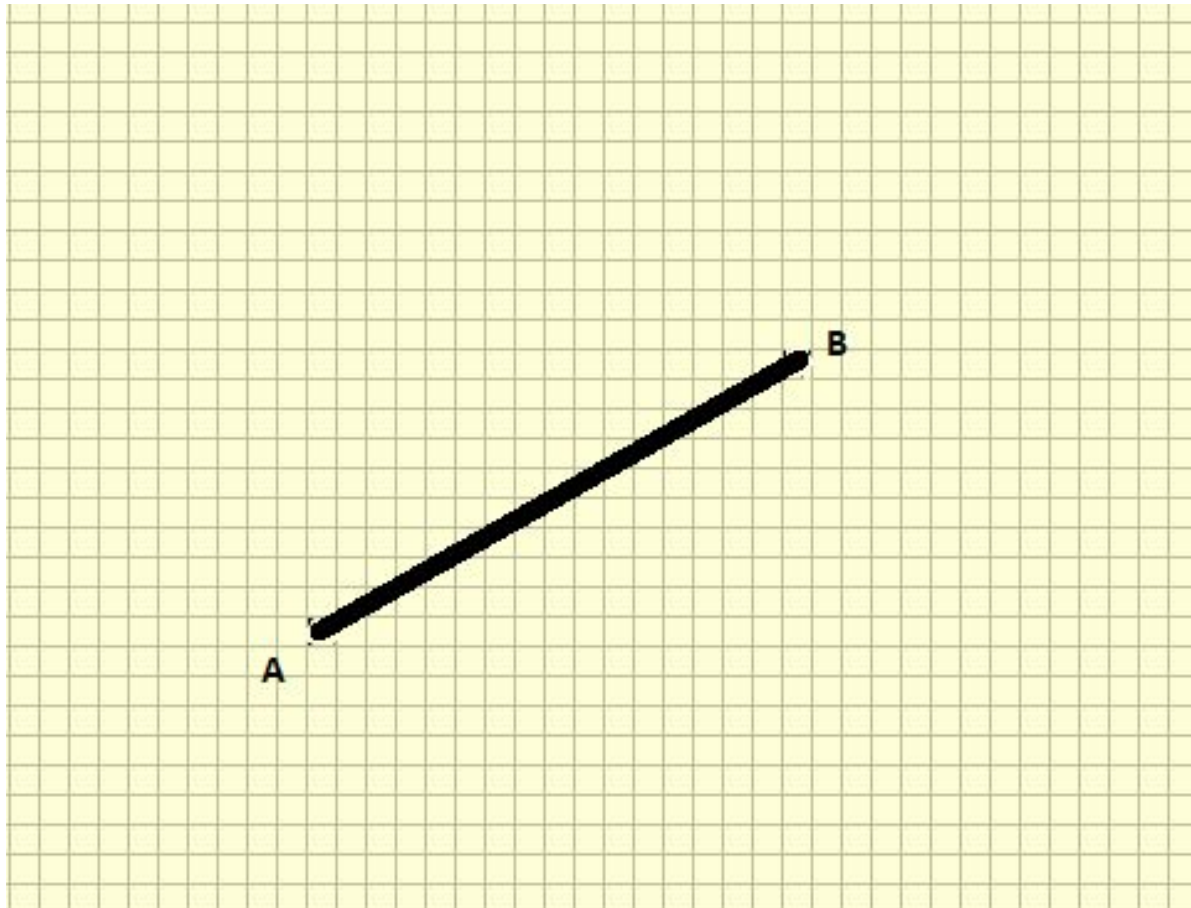
Some Questions (1)

We have two points namely A and B. How does computer draws the line AB?



Some Questions (1)

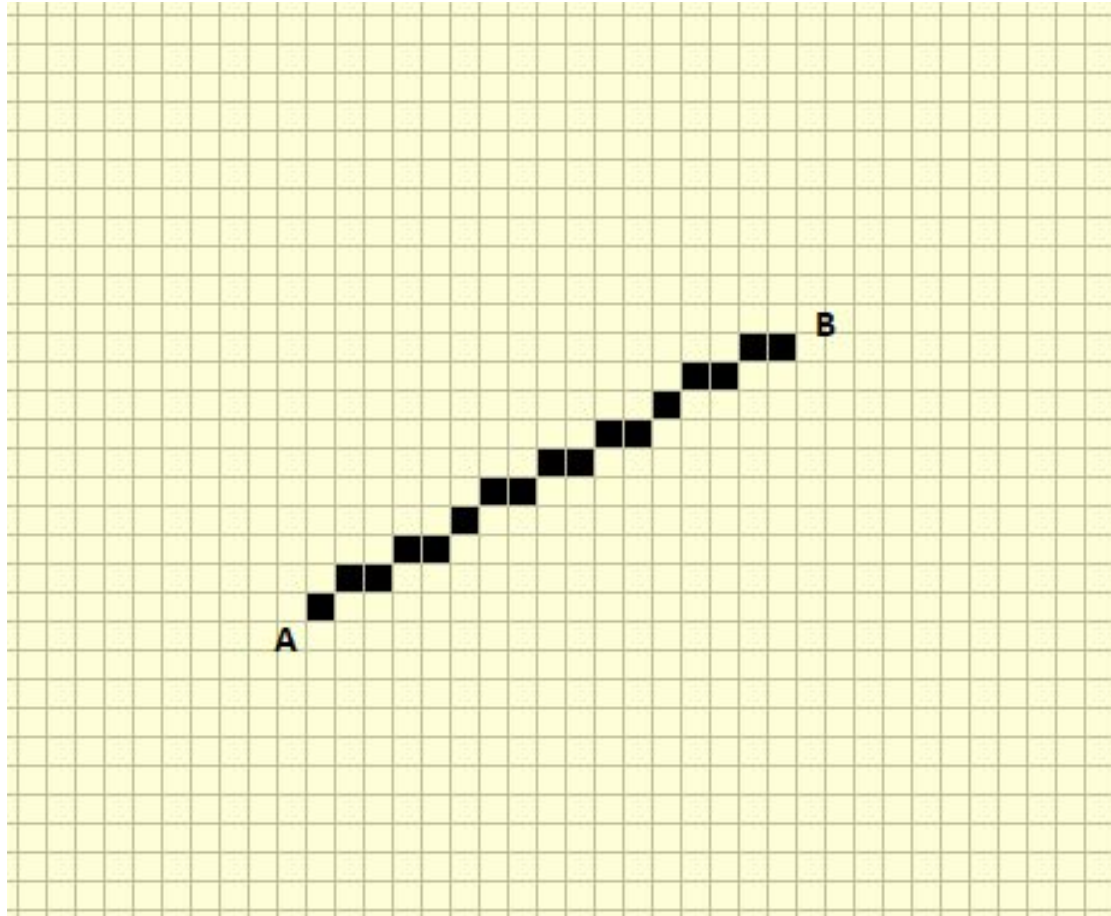
We have two points namely A and B. How does computer draws the line AB?



Is this correct?

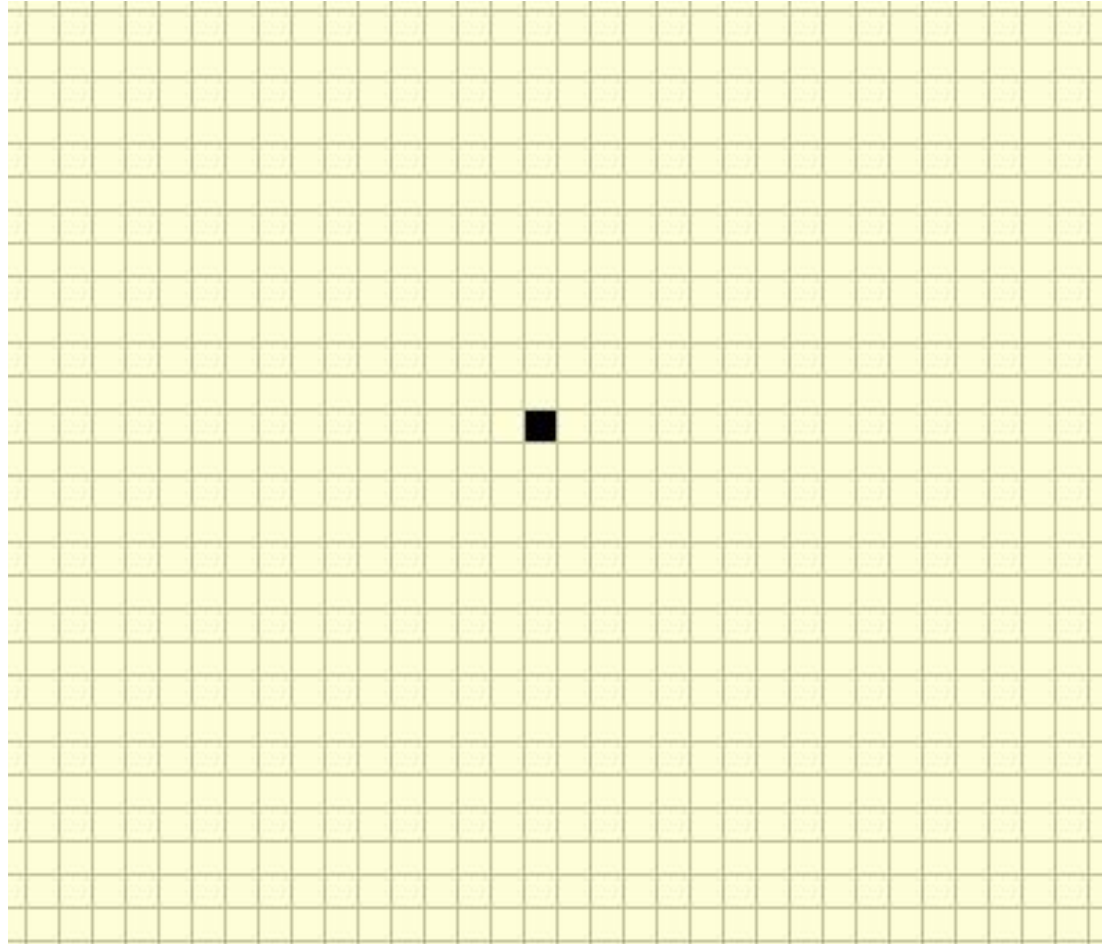
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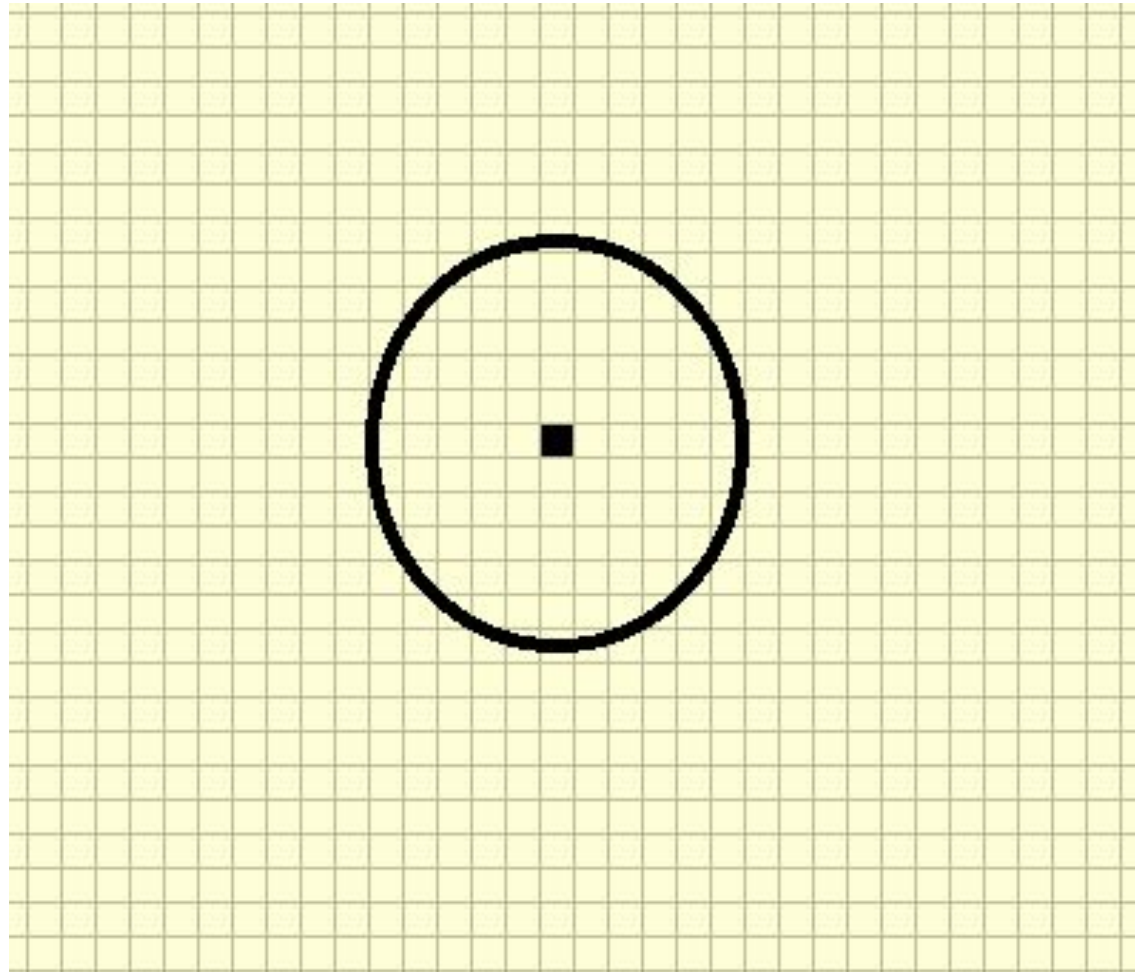
Some Questions (2)

How does computer draws a circle having radius of 5 pixels?



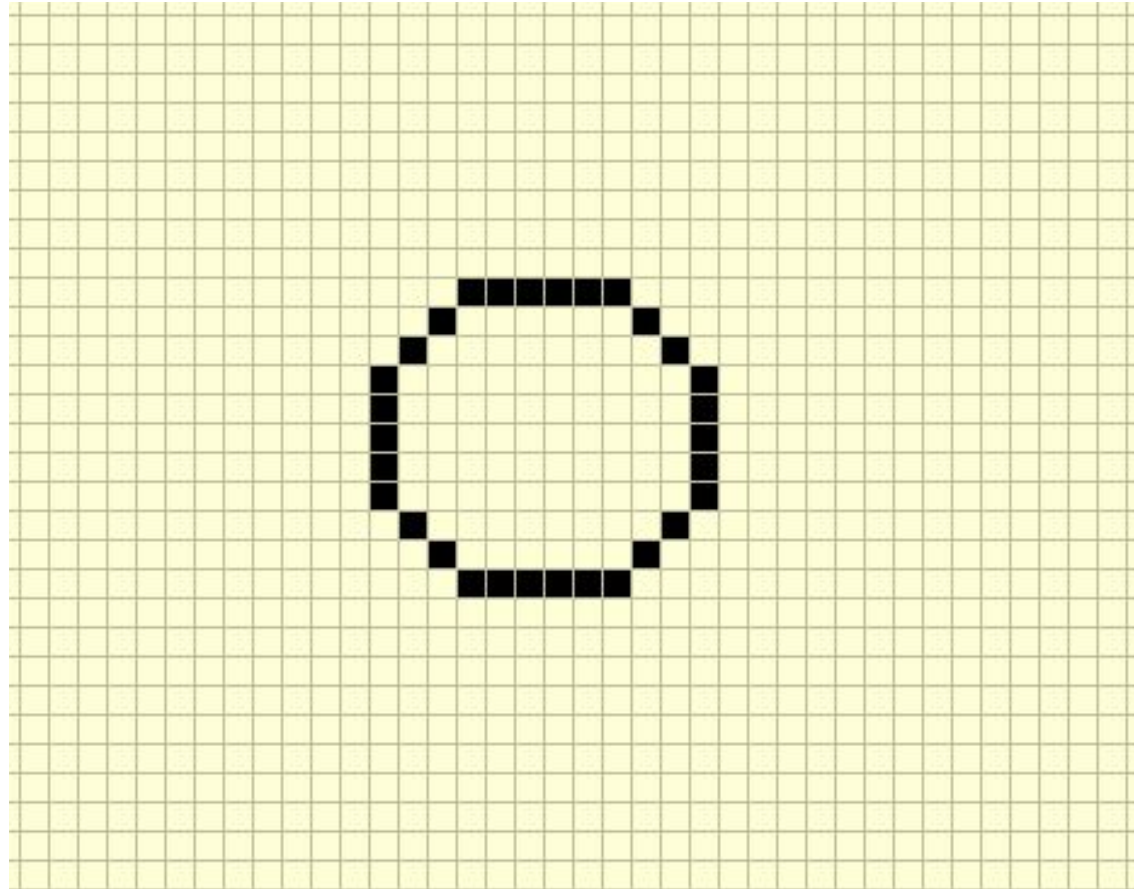
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How does computer draws a circle having radius of 5 pixels?



Some Questions (2)

How does computer draws a circle having radius of 5 pixels?



**A Pixel Is *Not* A Little Square,
A Pixel Is *Not* A Little Square,
A Pixel Is *Not* A Little Square!
(And a Voxel is *Not* a Little Cube)¹**

Technical Memo 6

Alvy Ray Smith
July 17, 1995